WORKFLOW CASE STUDY & TECHNICAL REPORT

Kombee Platform Development Workflow

From BRD/FRD Ingestion to Full-Stack Scaffold and Visual Modernization

Project: Kombee Assessment Platform

Date: October 2026

Framework: React 19 + TypeScript + Vite

Executive Summary

This document formalizes the multi-agent AI development workflow utilized to create the **Kombee Technical Assessment & Skills Platform**. The project utilized a sequenced pipeline combining **OpenAI ChatGPT** (for document comprehension and master prompt synthesis), **Anthropic Claude Code** (for architecture scaffolding and initial implementation), and **Google DeepMind Antigravity** (for UI/UX modernization, design system alignment, and zero-regression verification).

Workflow Pipeline Overview

OPENAI CHATGPT

Role: Analysis & Prompt Generation

- Ingested raw BRD, FRD & Brand Kit
- Structured product requirements
- Generated comprehensive agent prompt

CLAUDE CODE

Role: Architectural Scaffolding

- Scaffolded React 19 + TypeScript codebase
- Implemented full state & routing matrix
- Built 10+ core domain modules

ANTIGRAVITY

Role: UI Redesign & Delivery

- Modernized dashboard aesthetics
- Aligned with modern fluid SaaS UI
- Zero-regression build verification

1 Phase 1: Requirements Ingestion & Prompt Synthesis (ChatGPT)

The development process began by feeding the foundational enterprise documents into **ChatGPT**:

- **Business Requirements Document (BRD):** Outlining the business context of an enterprise skill assessment system, multi-tenancy needs, proctoring requirements, and role hierarchies (Platform Admin, Recruiter, Reviewer, Candidate).
- **Functional Requirements Document (FRD):** Defining specific system capabilities including timed evaluations, dynamic coding sandboxes, integrity monitoring, scorecard reviews, and analytics dashboards.
- **Brand Guidelines:** Defining the core palette centered around signature **Kombee Indigo (#4648D4)**, Poppins typography, and modern SaaS aesthetic values.

Outcome: ChatGPT synthesized these inputs into a robust, structured master implementation prompt designed specifically for autonomous code generation agents.

2 Phase 2: Architecture Scaffolding & Initial Codebase (Claude Code)

The generated master prompt was provided to **Claude Code** to generate the application codebase from scratch:

- Built a production-grade **React 19 + TypeScript + Vite + Tailwind CSS v4** application.
- Engineered end-to-end features: Multi-tenant auth simulator, Candidate Management, Question Bank, Assessment Builder, Reporting & Analytics, and Integrity Proctoring.
- Implemented clean mock API services, Zustand state stores, and robust route guards.

Handover Milestone: Upon completing the initial functional codebase, Claude Code credit limits were reached. The fully functional codebase was preserved and transitioned to Google Antigravity.

3 Phase 3: Visual Modernization & Design Polish (Antigravity)

Google Antigravity was brought in to elevate the user interface from a minimalist functional scaffold into an eye-catching, production-ready SaaS dashboard.

> **Design Direction & Client Iteration**
>
> Following user guidance and reference designs provided from high-end SaaS platforms (Intellecta / modern analytics), Antigravity preserved 100% of the underlying logic while overhauling the visual language to be fluid, modern, and aligned with the authentic Kombee brand guidelines.

UI Surface	Initial State	Antigravity Modernized State
Dashboard Header	Standard plain text greeting	Fluid greeting with inline telemetry summary (`Invited ↗`, `Avg completion ↗`) and pill CTA button (`+ New assessment`).
KPI Metric Cards	Basic uniform white boxes with text	Featured Hero Brand Indigo card (`#4648D4` gradient) with translucent pill tags, paired with soft white cards featuring pastel status badges.
Candidate Funnel Chart	Simple default vertical bars	Segmented rounded purple bars with stage progression pills (Invited, Started, Submitted, Passed) and floating hover tooltip.
Activity & Scheduling	Generic metric list	Timeline cards with soft pastel status pills (Lime, Lavender, Gray) matching contemporary UI patterns.
Navigation (Sidebar & Topbar)	Standard rectangular dark sidebar	Curved white dock with brand badge, glowing active pills, and floating pill search bar with circular avatar controls.
Login & Identity Hub	Simple form layout	Clean, light-themed authentication hub with tenant badges, user avatar indicators, and interactive launch actions.

4 Phase 4: Zero-Regression Verification & Delivery

Throughout the entire Antigravity design elevation phase, a strict **Zero-Regression Policy** was enforced:

- **Business Logic Preservation:** Zero modifications to routing, role-based access, tenant synchronization, mock APIs, or state stores.
- **Build Verification:** Production build executed cleanly with `npx vite build` in **664ms with 0 errors**.
- **Local Hosting:** Verified live running instances on **http://localhost:5173/** and **http://localhost:5174/** with active Hot Module Replacement (HMR).

5 Tooling & Contribution Summary

Stage	AI Platform / Tool	Primary Output
Analysis & Prompting	OpenAI ChatGPT	Synthesized BRD/FRD/Brand requirements into an actionable agentic prompt.
Core Implementation	Anthropic Claude Code	Generated complete React + TypeScript architecture and 10+ feature modules.
UI Modernization & Polish	Google Antigravity	Transformed UI into a fluid, eye-catching SaaS dashboard matching brand standards.